

Student Answer Sheet

Student: Sample Student A

Q1

Photosynthesis is the process by which green plants make food using sunlight, carbon dioxide and water. Sunlight provides the energy needed for this process.

Q2

Osmosis is the movement of water through a semi permeable membrane from a region of higher water concentration to lower water concentration. For example raisins swell when kept in water.

Q3

Newton's first law says an object remains at rest or in uniform motion unless acted on by an external force. Second law relates force, mass and acceleration. Third law says every action has an equal and opposite reaction. A balloon moving forward as air escapes is an example of the third law.

Q4

Evaporation happens at the surface of a liquid and can occur at many temperatures. Boiling happens throughout the liquid at its boiling point.

Q5

The sky looks blue because sunlight contains many colours and blue light is scattered more by the particles in the atmosphere than most other visible colours.

Q6(a)

Renewable energy comes from sources that are naturally replenished. Examples are solar energy and wind energy.

Q6(b)

Solar energy is environmentally friendly because it produces electricity without burning fuel and therefore produces very little direct air pollution during operation.

Q7

Plant cells have a cell wall and chloroplasts, while animal cells do not. Plant cells generally have a large central vacuole, while animal cells have smaller vacuoles.

Q8

The water cycle is the continuous movement of water between the atmosphere, land and oceans. Its main stages include evaporation, condensation and precipitation.

Q9

The heart is a muscular organ that pumps blood throughout the body. It sends oxygen-rich blood to body tissues and returns oxygen-poor blood to the lungs for oxygenation.

Q10

A food chain shows how energy and nutrients pass from one organism to another through feeding. Example: grass, grasshopper, frog, snake, eagle.